

Assignment 1

1.

6

The instructions CPU1 and CPU2 support can be divided into four classes according to their CPI (class A, B, C, and D). CPU1 has a clock rate of 2.7 GHz and the instruction classes have CPIs of 7, 2, 3, and 6 respectively whereas CPU2 has a clock rate of 3.0 GHz and the instruction classes have CPIs of 5, 4, 2, and 1 respectively. Now suppose, a program has an instruction count of 10^6 , and the instructions are divided into classes as follows:

- 20% class A,
- 50% class B,
- 20% class C, and
- 10% class D.

Now answer the following questions:

1. Calculate how many more clock cycles per instruction on average does the CPU1 take compared to CPU2?
2. Calculate the difference between the execution time in these two CPUs

2.

4

CPU1 has a clock rate 4 GHz and another CPU2 has clock rate 2GHz. The CPI of a program in cpu1 is 4 and in cpu2 is 6. Calculate which cpu performs better for this program in ips?